FEDERICO ALBERTO ASTOLFI

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SUMMARY

I like to inspect things from close and to fashion new ways to fathom them. My education in mathematics has given me the language to formalize and the patience to model. My years in high-school and my free-time readings have nurtured in me the passion and the curiosity for language and its expressions, and I carry within me this humanistic side to sharpen both my understanding of things and its outward communication.
In few words, I am a **Quantum Scientist** with an heterogenous education whose unmatched enthusiasm is easily triggered by challenges and whose energy is not exhausted by hurdles. My strong quantitative skills developed in years of study are accompanied by my qualitative skills; together with my ability to adapt to and learn from different frameworks, this mix makes me a solid asset that you want to have in your portfolio.

WORK EXPERIENCE

Ph.D. Student, Université de Strasbourg, Strasbourg, France — **Oct 2024 - Present**

- Employed at Centre Européen des Sciences Quantiques (CESQ) and Institute of Supramolecular Science and Engineering (ISIS).
- Funded by **Marie Skłodowska-Curie Actions** (MSCA) doctoral network "**Machine Learning for Quantum**".
- Working with Quantum Computing, with a focus on **Quantum Error Correction** with **ML** and **Quantum Optimal Control**.

Intern, Leithà - Unipol Group, Bologna, Italy — **Apr 2025 - Jun 2025**

- Investigated techniques of hyperparameter tuning for **Neural Networks**.

Test Engineer, TXT Group, Cologno Monzese, Italy — **Dec 2023 - Aug 2024**

- Engaged in software testing for aero-machinery projects for the client Leonardo Helicopters.
- Involved in the design and execution of test plans to ensure the quality, functionality, and performance of software products.

Swimming teacher and Lifeguard, A.S.D. Asa, Cinisello Balsamo, Italy — **May 2018 - Dec 2023**

- Responsible of the improvement of the aquatic skills, with expertise ranging from the building of water confidence in both kids and adults to the refinement of the swimming technique.
- In charge of the safety of the swimmers, through surveillance of the pool and control of the water parameters.

EDUCATION

Ph.D. programme in Physics **Oct 2024 - Present**

Université de Strasbourg, CESQ & ISIS

- Applying techniques from Optimal Control Theory to quantum systems for faster gate implementation.
- Employing Machine Learning to advance the existing protocols for the decoding of quantum error correcting codes.

Master of Science in Mathematics **Sep 2021 - Feb 2024**

Università Statale di Milano

- Erasmus experience of six months at Rheinische Friedrich Wilhelms Universitaet in Bonn, Germany.
- Final thesis: “Minimal Lagrangian orbits of Hamiltonian actions over KE and KRS”.
- Final evaluation: 110/110 cum laude.

Bachelor of Science in Mathematics **Sep 2018 - Jul 2021**

Università di Milano Bicocca

- Final thesis: “Una introduzione alla teoria di Morse”.
- Final evaluation: 110/110 cum laude.

Diploma in Classical Studies **Sep 2013 - Jun 2018**

Liceo Classico Giulio Casiraghi

- Final evaluation: 100/100.

HARD SKILLS

- **Programming Skills:** Python, Julia
- **Languages:** Italian (native speaker), English (proficient and confident user), French (lower intermediate)

SOFT SKILLS

- Efficient in communication
- Team player
- Critical thinker
- Good listener
- Proactive
- Resilient